What is a Business Decision

What is a Operational Business Decision?

A large part of conducting any business is making decisions. Some of these are strategic: should we enter a certain market, how should we design our new product, which partners and distribution channels should we choose? Others are more routine, made manually or automatically during everyday business operations. The latter are *operational* decisions.

Operational decisions are determinations that businesses make on a regular basis, a selection or calculation of an outcome that depends on a number of prevailing circumstances (inputs) and which, ultimately, has an observable impact on the behavior of an organisation. They include making determinations like:

- Should we extend a line of credit to this customer? On what terms?
- Should we initiate an inquiry into a customer's insurance claim or just pay it?
- What products should we recommend to a client when they visit our website, given their past behavior?
- Does this trade fee structure satisfy compliance regulations?

Many operational decisions need to happen many times a second and so they are often automated.

Why are Decisions Important?

Operational decisions determine the day-to-day profitability of the business, how effectively it retains customers or how well it manages risk. Often the quality and consistency of decision-making determines your client reputation—for some clients their sole perception of your company is obtained from the outcome of these decisions. Operational decisions may not seem as important as the strategic ones, but they are more voluminous and their significance eventually adds up. Decisions are often of special importance in regulatory compliance applications. Digital transformation is partially about how this decision-making can be made more customer focused.

The logic of some decision-making is intellectual capital: it helps to establish or maintain a competitive advantage for your company—it represents what you do to better your rivals, your unique selling points. You need to identify the important decisions you are making, define the decision-making process transparently, ensure they are made accurately, monitor their performance and manage their evolution and improvement. This process of explicitly managing your decisions is business decision management. The means of transparently representing decisions is decision modeling.

Business Decisions and Business Rules

Business decisions are necessarily coarser grain and more tangible to business subject matter experts that business rules. You should be careful not to confuse decisions and rules.

Why Decision Management?

Do you ever have the feeling that running your business is rather like being on the prow of an ocean liner at full speed in a squall? You may have a feeling of progress, your engine-room may be generating considerable power, but ultimately: your route to the destination is not clearly visible, you cannot discern if the ship is tracking the right course towards that destination and you know that it would be difficult to change course quickly to avoid an obstacle if the need arose. Better, perhaps, to be in the bridge of a fully-equipped motorboat where you have charts that clearly define your destination and your route to it, a GPS which shows you how you are tracking that route and the ability to change course quickly and take advantage of ever changing prevailing circumstances.
To control a business effectively you need:

- To have an cogent strategy (a route to a specific destination),
- To understand how well you are currently pursuing this course (**performance measurement**)
- To grasp *precisely why* this is the case (**transparency**) and
- the ability to change it quickly if it proves to be ineffective (**agility**).

Decision Management (DM) is a technique and a discipline that provides the latter three capabilities: transparency, agility and performance measurement.

Transparency

DM supports transparency by allowing you to gain a rich and precise understanding of your (current) operational business decisions – the small high volume choices your business makes every day when dealing with clients: *should I loan money to this client? What should the upper limit for the loan be? How much collateral is required? What should I charge a client for this service?* This understanding is achieved through modelling your business logic, either from scratch, or using an existing implementation as a start point.
Much as a data architect might use data modelling to understand and improve how data is sourced, used and generated by an enterprise, a decision architect uses Decision Modelling to capture and enhance the effectiveness and integrity of business logic. Logic can be modelled using a mixture of simple decision tables and complex analytics. The process of capture makes business logic *explicit* and visible for all to see. The crucial consequences of this process are:

- that your business logic is liberated from IT infrastructure and implementation detail (or the heads of individual experts) and, instead, expressed plainly in business terms so that it can be reviewed, critiqued and improved by subject matter experts;
- your business logic is simplified by normalization (rather like data is normalized when expressed in third normal form) making it even easier to understand and more self-consistent;
- the dependencies between different aspects of your business decisions are explicitly articulated: you can now understand how each policy impacts others and check their internal consistency; and
- Modelling prevents loss of knowledge by staff attrition and ad-hoc documentation.

Agility

Once you have a thorough understanding of your business logic and policies, DM supports agility through automation and governance. When these decisions are automated and their inter-dependencies are

understood, they can be controlled more directly (by subject matter experts) and changed much more quickly and accurately. The usual ‘ceremony’ of release can be greatly streamlined. As automation provides the speed, DM’s strong governance process provides the safety by ensuring that all decision modifications are reviewed, authorized and their impact is as well understood as is possible ahead of release. Simulation is used to articulate the full impact of proposed policy changes before any commitment is made. Traceability ensures that the detailed rationale of all automated outcomes can be ‘explained’ by the automated system and used to define changes. DM provides the framework that maximizes the effectiveness of simulations and traceability.

Performance Measurement

Providing your business decisions are defined within the context of a business process, you can use DM to support a programme of continuous performance measurement and improvement. Defining metrics for your process enables you to continuously measure how effective your decisions are and highlights opportunities for change. Defining a business process to utilize your business decisions also highlights opportunities for reuse of decisions – promoting greater consistency across the process itself and amortizing implementation effort.

Better than Business Rules

It may be that you have heard of, or use, business rules to achieve some of these goals. By making a direct link to business goals and being explicit about dependencies, Decision Management goes one step further and is demonstrably more enterprise scalable and effective that a Business Rule only approach.

The Proof of the Pudding

The real benefits of DM can best be appreciated by practical examples. DM is used today by retail and investment banks, insurers, brokers to:

- Target cross selling of their products
- Price products and assess risk
- Dispatch their KYC and customer eligibility checks
- Guide user interaction with their channels (e.g., intelligent web site navigation)
- Implement channel routing and constraint enforcement
- Ensure (and prove) their compliance to industry regulations (e.g., capital/collateral adequacy, Basel III, withholding tax, volatility buffering)
- Detect and handle fraud

DM is effective in any scenario that involves the need to understand, manage and change non-trivial business logic for competitive advantage.

If you are interested in the benefits of DM and would like to understand them better in the context of a detailed case study please contact us to arrange a review.

Decision Modeling: The Bottom Line

by Jan Purchase | Aug 29, 2017 | Experience Report | 0 comments

Candidate Eligibility for Loan				
U	Candidate Salary	Candidate Credit Rating	Candidate Employment History	Candidate Eligibility for Loan
		AAA, AA, A, B, C, D	EXCELLENT, GOOD, INTERMEDIATE, POOR, NONE	APPROVE, REVIEW, REJECT
1	>= 50000	AAA , AA , A	EXCELLENT , GOOD	APPROVE
2			INTERMEDIATE	REVIEW
3			POOR , NONE	REJECT
4		B	EXCELLENT	APPROVE
5			GOOD	REVIEW
6			INTERMEDIATE , POOR , NONE	REJECT
7		C	EXCELLENT	REVIEW
8			not(EXCELLENT)	REJECT
9		D	-	REJECT
10	<50000	-	-	REJECT

Why should organizations model their important business decisions as part of digital transformation? We've been asked so many times to explain how our clients have benefited from decision modeling that we decided to capture it here. This article covers seven reasons to adopt decision modeling and summarizes the bottom-line benefits decision modeling has brought to companies that use it effectively.

The Importance of Decisions

A large part of conducting any business is making decisions. Some of these are strategic: should we enter a certain market, how should we design our new product, which partners and distribution channels should we choose? Others are more routine, made manually or automatically during everyday business operations. These are *operational* decisions.

Operational decisions include answering questions like:

- Should we extend a line of credit to this customer? On what terms?
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- What products should we recommend to a client when they visit our website, given their past behavior?

Some operational decisions need to happen many times a second and so they are often automated. These decisions determine the day-to-day profitability of the business, how effectively it retains customers or how well it manages risk. Often the quality and consistency of decision-making determines your client reputation—for some clients their sole perception of your company is obtained from the outcome of these decisions. The logic of some decision-making is intellectual capital: it helps to establish or maintain a competitive advantage for your company—it represents what you do to better your rivals, your unique selling points. You need to make these decisions accurately, monitor their performance and manage their evolution. Digital transformation is partially about how this decision-making can be more customer focussed.

Part of a business analyst's job is to identify, define and improve these decisions. That's a challenge because often the details of the decision-making approach are trapped in the heads of subject matter

experts, in reams of obscure documentation, in spreadsheets or within the code of software systems. These decisions may already be automated, but they are not leveraged because few people understand how they work, how effective they are or how to change them.

What is Decision Management?

Decision Management is a means of explicitly identifying and describing your business's operational decisions—much as you would any other vital business asset (like data or process)—this allows you to 'bring them out of the shadows'. It is an approach (and a technology stack) for describing, sharing, managing change and checking decision performance to see how they are contributing towards your enterprise goals. The goals of managing decisions are:

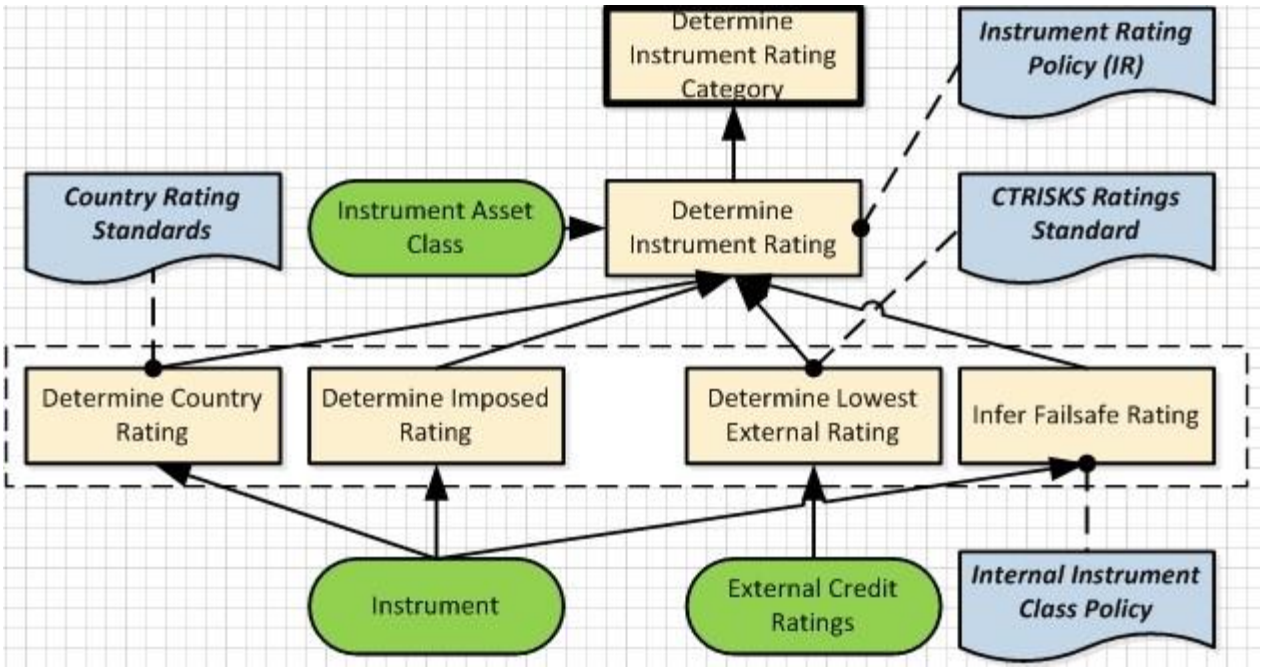
- To improve the **transparency** of decision-making by representing decisions in a format that allows business experts and IT to share a deep understanding of them.
- To increase the **productivity** of decisions by automating them in a disciplined way to increase throughput and reduce delays.
- To enhance the **accuracy** of decisions by supporting monitoring to measure their business effectiveness and collaboration to improve their definitions.
- To boost business **agility** and **adaptability** by ensuring that decisions are easy to change and that the full consequences of such change can be quickly understood. The fact that decision models are **executable** without need for translation into code further improves **agility**.
- To improve the **consistency** of business operations by ensuring that regardless of business process or channel, the same decisions are reused where appropriate.

An important part of this is *modeling* decisions.

What is Decision Modeling?

Decision Modeling (DM) is a part of decision management that focuses on representing business decisions using a standardized notation designed for use by business analysts and subject matter experts, rather than developers.
Through Decision Modeling, business analysts can build and share robust documentation of how their business decisions work. A decision model consists of two views: the Decision Requirements View and the Decision Logic View.

The Decision Requirements View describes how complex decisions are decomposed into simpler sub-decisions and exactly what data and business knowledge are required to support them. It shows the hierarchical structure of a decision.
An example is shown on the right. The decision (the yellow rectangle with a thick border) is broken down, showing its dependencies on sub-decisions (yellow rectangles), input data (green roundrects) and knowledge sources (blue rectangles with curvy bases).
Decision requirements diagrams are a powerful way of showing the integration of business rules, analytics, optimization models, business know-how and source data within an architecture. They facilitate all of these by providing a context for combining them into an integrated whole. Subject matter experts find that creating this view is an effective way of eliciting and consolidating their understanding of key decision-making.



Failsafe Rating			
U	Issuer Type	Instrument Class	Failsafe Rating
	<Issuer Type>	<Instrument Class>	<Instrument Rating>
1	*US MORTGAGE SECURITIES*	-	*AAA*
2	not(*US MORTGAGE SECURITIES*)	*US AGENCY, US TBILL*	AAA
3		*AP AGENCY*	*AA*
4		*FAR EAST AGENCY*	A
5		*GOVT EMERGING*	*AA-*
6		*UNKNOWN*	*UNKNOWN*

The Decision Logic view (see example on the left) defines the logic of decisions. Often, but not always, it uses Decision Tables to represent how decisions determine their outcomes. This allows business analysts to describe exactly how decision-making is done in their organizations. Decision models can also represent logic in other formats (e.g., decision trees, text) or integrate with other means to make a decision (e.g., analytics, cognitive models and optimization engines).

Decision Models are not just about requirements. The precision of the decision logic view means that decision models can be directly *executed*. Therefore they can be tested against real business data and even deployed into productions systems without the need for developers to translation them into code. This model-driven approach to decision architecture reduces errors and allows for rapid change in decision-making—it provides *safe agility* for business decision-making and ensures that your specification of decision-making is always specific and consistent with the implementation.

Seven Reasons to Model Your Business Decisions

1 Manage Complexity, Maintain Integrity

Unlike ad-hoc spreadsheets and even Business Rules, Decision Models are designed to scale effectively while remaining easy to understand. They achieve this by ‘divide and conquer’: separating complex problems into logical hierarchies of simpler concepts which can then be expressed and evolve

independently, but yet collaborate explicitly and effectively.
A Decision Model is explicit about decision requirements: logical dependencies between business decisions and: other (sub) decisions, externally-defined policies, regulatory mandates, laws and business data. This makes it easier to determine the exact impact of changing decisions, data and externally defined policies. Many DM tools can illustrate the impact of making an internally- proposed or externally mandated change, for example a change in data standards or a regulatory compliance mandate.
Furthermore, DM gathers all related logic into one place: promoting logical integrity within rule sets, helping to identify gaps and inconsistencies within the logic as requirements are being formulated and thereby reducing errors from the start.
The explicit statement of dependencies engendered by decision modeling helps to identify real data requirements very quickly, avoiding over-provision of expensive data and the loss of productivity associated with late discovery of missing data.

2 Communicate Business Policy Transparently and Precisely

One of decision modeling's principle motives is to capture and communicate business decisions in a way that's transparent to business subject matter experts. This allows them to understand, verify and even simulate decisions. DM provides a business-oriented representation that is separate from implementation details, yet precise enough to support execution. Explicit representation of business decisions liberates them from the bowels of IT systems and from the heads of subject matter experts, bringing them out into the light for all to see and avoiding loss of business expertise caused by legacy system decommissioning and staff attrition.
DM aims to increase the effectiveness of communication between business analysts and development teams. Decision definitions can be unambiguously shared and collaboratively reviewed so that fewer mistakes are made and any mistakes are rapidly identified and resolved. Time can be spent more productively in discussing the correctness and ramifications of requirements rather than resolving misunderstandings about their meaning. This time saving radically decreases time-to-market and error rates.
Decision models are also a very effective platform for training new staff in organizations' decision-making logic.

3 Enable Rapid, Model-Driven Development

Decision models can define logic so precisely that they are *executable*. This is crucial because it means that these models are not just boxes, lines and text but the basis of a testable and deploy-ready decision service. Executable decision models can become the prime record for decisions—making them a *living* specification of operational decision-making. They eliminate the need for IT to perform an error-prone translation of decision-making practices from dusty specification documents arcane spreadsheets and interviews with experts into a production system. Instead the decision model that business analysts and IT develop together can be simulated, tested and deployed directly.
Decision models can be associated with metadata that defines the true business value of a decision. For example, the business goals the decision is contributing to and how well it performed (e.g., how much money was saved, how much quicker was an outcome reached). This means when they are executed this performance can be continuously monitored and any undesirable trends quickly identified and addressed.

4 Support Agile Change Management

Explicit definitions of business decisions that can be understood and simulated allow business analysts to take direct control of the evolution of business policy, rather than having to intermediate through IT teams that own the systems that implement this policy. Analysts can correct mistakes and make improvements with little IT involvement (using simulation and impact analysis), deploying changes to a integration testing platform for testing by IT only when they are sure they are correct.
This thorough understanding of decision dependencies enables effective change impact assessments and leads to agile change cycles. These advantages cannot be provided by existing approaches like Business Rules.

5 Render Decision-Making Fully Accountable

DM and business rules together support complete traceability from policy mandate to production system, allowing every outcome reached by a decision to be accompanied by a justification (in terms of the policy mandate) which can be examined in hindsight. This can assist in meeting regulatory requirements by demonstrating compliance with a regulation and help to diagnose faults quickly. his traceability can also be used to analyze the effectiveness of decision-making over time and suggest innovations for improvement.

Such accountability can also streamline change management when regulatory mandates are updated because the parts of the mandates that change can be readily mapped to the parts of the decision model that support them and therefore need to be amended.

6 Achieve Standardization

DM standardizes the format of business operational decision requirements using the Decision Model and Notation standard (DMN). DMN is an open standard defined, ratified and published by a team of experienced vendors under the auspices of the Object Management Group Inc. (OMG). The OMG is an international, not-for-profit, technology standards consortium specializing in the development and custodianship of standard notations used by businesses in over twenty market verticals. Many of their standards are already well known to business analysts (e.g., BPMN, UML).

DMN provides a powerful and flexible notation and formal grammar for representing decision models and business logic. It has a simple, extensible and easy to learn structure which captures much of the essential information required to define decisions. It is designed for use by business analysts and business subject matter experts as well as IT professionals.

The use of such a standard means every modeler uses the same means to represent business decisions—so they can understand each other's work even if they don't have the same background. This is not only a big improvement over the 'everyone has their own spreadsheet format' free for all but also allows companies to hire new DMN-aware BAs in the knowledge that they will understand the models already created by others. This increases transparency: allowing effective review of business requirements by all stakeholders across the world, not just those in the inner team. It also promotes consistency between different decision authors, different teams and over time.

7 Simplify Other Models

Frequently, aspects of decision-making are scattered across a range of text or spreadsheet documentation, hidden in business processes or hinted at by data models. By explicitly modeling decisions using a separate, standard model which integrates with business data, business process and business motivation models, decision modeling reduces the complexity of these other models, allows each to express those aspects of a system to which they are best suited and allows each to evolve at their own rates. DMN is designed to interoperate with BPMN and UML and is ratified by the same standards organisation.

The Bottom Line

The bottom line is that Decision Modeling, by providing a transparent, traceable, scalable and precise representation of business decisions that is executable, that integrates analytic, optimization and cognitive technologies, that integrates decision making within a business process and that tests the dependencies that decisions have on business data and external authorities:

- **Lowers time to market**, by allowing business analysts to build, test and improve transparent models of decision making without specialist infrastructure or heavy IT involvement
- **Lowers costs** and improves reliability not only by supporting automation of decision making, but also by avoiding logical flaws and data provision errors
- **Improves agility** by allowing the full impacts of proposed changes to be seen before they are committed and by facilitating rapid amendments

- **Manages complexity** by supporting the hierarchical decomposition of processes and integrating many different logic representation styles into a coherent whole
- **Improves business engagement** by ensuring early and constant involvement by subject matter experts and business analysts in the evolution of business decisions
- **Supports the continuous improvement of business operations** by allowing the measurement of decision effectiveness and making them accountable for their consequences.

Find Out More About the Practical Applications of Decision Modeling

Decision modeling is most frequently used in areas where high operational throughput, complexity or change are expected. Business areas where accountability and effectiveness is important also benefit. These include but are not limited to: risk assessment and management, pricing, regulatory compliance, KYC, eligibility, cross-selling, telecommunications tariffs and fraud detection.

What is Business Intelligence?

BI(Business Intelligence) is a set of processes, architectures, and technologies that convert raw data into meaningful information that drives profitable business actions.It is a suite of software and services to transform data into actionable intelligence and knowledge.

BI has a direct impact on organization's strategic, tactical and operational business decisions. BI supports fact-based decision making using historical data rather than assumptions and gut feeling.

BI tools perform data analysis and create reports, summaries, dashboards, maps, graphs, and charts to provide users with detailed intelligence about the nature of the business.

Four types of BI users

Following given are the four key players who are used Business Intelligence System:

1. The Professional Data Analyst:

The data analyst is a statistician who always needs to drill deep down into data. BI system helps them to get fresh insights to develop unique business strategies.

2. The IT users:

The IT user also plays a dominant role in maintaining the BI infrastructure.

3. The head of the company:

CEO or CXO can increase the profit of their business by improving operational efficiency in their business.

4. The Business Users"

Business intelligence users can be found from across the organization. There are mainly two types of business users

1. Casual business intelligence user
2. The power user.

The difference between both of them is that a power user has the capability of working with complex data sets, while the casual user need will make him use dashboards to evaluate predefined sets of data.

Advantages of Business Intelligence

Here are some of the advantages of using Business Intelligence System:

1. Boost productivity

With a BI program, It is possible for businesses to create reports with a single click thus saves lots of time and resources. It also allows employees to be more productive on their tasks.

2. To improve visibility

BI also helps to improve the visibility of these processes and make it possible to identify any areas which need attention.

3. Fix Accountability

BI system assigns accountability in the organization as there must be someone who should own accountability and ownership for the organization's performance against its set goals.

4. It gives a bird's eye view:

BI system also helps organizations as decision makers get an overall bird's eye view through typical BI features like dashboards and scorecards.

5. It streamlines business processes:

BI takes out all complexity associated with business processes. It also automates analytics by offering predictive analysis, computer modeling, benchmarking and other methodologies.

6. It allows for easy analytics.

BI software has democratized its usage, allowing even nontechnical or non-analysts users to collect and process data quickly. This also allows putting the power of analytics from the hand's many people.